Predicted densities and weights in the historical CH instrument

1. Put the historical information on modern county units. Let P_c be historical population assigned to modern county c by geographic overlap. Let n_c and a_c be modern employment and land area in square kilometres. Counties combined in the employment data are kept together. Two historical density definitions are available: ℓ_c is either the population-weighted mean of historical source log densities within c, or $\log(P_c/a_c)$. The first preserves historical source densities. The second spreads assigned population over the full modern county land area. These are different measures.

2. Predict modern employment density from counties. For each historical year and territory scope, estimate by ordinary least squares

$$\log(n_c/a_c) = \alpha + \pi\ell_c + X_c'\gamma + v_c, \qquad \widehat{m}_c = \widehat{\alpha} + \widehat{\pi}\ell_c + X_c'\widehat{\gamma}.$$

Here X_c contains the selected geography and transport variables. The regression uses eligible counties across the contiguous United States, an intercept, and no fixed effects or human-capital terms. This is a prediction of employment density, separate from the county productivity regression used to adjust the modern CH index. The fitted value $\widehat{m}_c$ and predicted employment $\widehat{n}_c = a_c \exp(\widehat{m}_c)$ do not depend on θ.

3. Choose what the prediction replaces. For a modern state g, the three alternatives are

$$z_g^A = \frac{\sum_{c\in g} P_c \widehat{m}_c}{\sum_{c\in g} P_c}, \qquad z_g^B = \frac{\sum_{c\in g} \widehat{n}_c \ell_c}{\sum_{c\in g} \widehat{n}_c}, \qquad z_g^{A+B} = \frac{\sum_{c\in g} \widehat{n}_c \widehat{m}_c}{\sum_{c\in g} \widehat{n}_c}.$$

A averages predicted log employment density with historical population weights. **B** keeps historical log density but gives counties weight according to predicted employment. **A+B** uses predicted density and predicted employment weights. All three use the same counties; weights sum to one within each state. The original historical option can retain more counties when prediction inputs are missing.

4. Coverage and estimation. Prediction requires positive historical population, a defined density and complete selected geography. Estimating the county regression also requires positive modern employment. Coverage counts known historical area in contributing counties, divided by the state's full mapped area. Zero-population counties cannot enter this prediction, although they count toward the original historical index's coverage. Choose 80% (the default), 95%, or no cutoff for both historical years. This filters states after the national county regression. Without a cutoff, small overlaps can admit a state; predicted weights use full modern county land area.

The prediction uses modern employment, so these are estimated instruments. Reported standard errors and tests condition on the fitted prediction coefficients. They do not include uncertainty from estimating those coefficients or establish the exclusion restriction.